

ResQGrid AI -- Gate 1 Supporting Evidence Dossier

Stakeholder field interviews · user survey analysis · empirical disaster citations

Purpose: This dossier provides primary and secondary validation backing up all problem claims in the Gate 1 Write-up. It contains full qualitative notes from five field conversations with named operational roles, quantitative results from a 32-respondent survey administered 22-25 Sep 2026, and verifiable citations from official disaster reports.

Part 1: Primary Field Conversations (Route A)

Conducted 21-25 September 2026. Semi-structured, 28-45 minutes each, recorded or noted with permission. Respondents are identified by operational role.

Conversation 1 -- Former Divisional Secretariat Disaster Management Coordinator, Gampaha District

42 min · In-person · Kelani / Attanagalu Oya basin

- **Context:** 8+ years coordinating localised monsoon flood response.
- **Intake overload:** "During the peak 48 hours of flooding, our phone lines are completely jammed. 8 people might be calling about the exact same two families on a rooftop. Because there is no deduplication or live spatial plotting, we spend hours writing names on paper pads and trying to figure out which Grama Niladhari division they belong to."
- **Resource blindness:** "We don't know where the Navy boats or Army trucks are once they leave camp. If a boat rescues 6 people and drops them off, we don't know they are free to pick up an elderly patient 400 metres away unless the coxswain calls us on his personal phone, which often has a dead battery."
- **Verdict:** "An automated system that turns WhatsApp messages into typed incidents on a live map and ranks urgency would cut our coordination time by more than half."

Conversation 2 -- Sri Lanka Navy RABS Boat Coxswain / First Responder

35 min · Phone interview · Western Province operational sector

- **Context:** Flood evacuations in Kolonnawa, Wellampitiya, Sedawatta during 2016 and 2025.
- **Blind routing:** "The single biggest danger is entering floodwater blindly. Water covers everything -- submerged iron fences, live fallen electricity lines, open drainage canals. Dispatch sent us down a road that was passable an hour ago but was now completely choked with debris. If the dispatch system knew that road was blocked and rerouted us via an alternative ridge road, it would save lives and prevent boat damage."
- **Mobile interface need:** "We cannot carry delicate laptops on a wet dinghy. Responders need a rugged, simple mobile screen that shows where to go, how many people are waiting, and lets us tap On Scene and Completed."

Conversation 3 -- 1990 Suwa Seriya Emergency Medical Technician (EMT)

30 min · Virtual interview · Colombo District

- **Context:** 5 years front-line pre-hospital emergency care during urban floods.
- **No clinical triage in dispatches:** "General callers just scream that water is entering their house. They don't know how to explain triage. We often dispatch an ambulance to find someone who is uncomfortable but safe on an upper floor, while three blocks away an insulin-dependent diabetic or oxygen-dependent elder is running out of supplies because the dispatcher had no way to extract that medical urgency from the call narrative."
- **Explainability needed:** "If an algorithm gives an incident a 90/100 score, I need to see exactly why -- e.g. Elderly + Hypothermia Risk. If it's a black box, doctors and dispatchers will never trust it."

Conversation 4 -- Community Flood Relief Volunteer Lead, Kolonnawa Relief Network

28 min · Semi-structured interview

- **Context:** Coordinated boat rescues across 14 Grama Niladhari divisions, 2016 and 2025.
- **WhatsApp group breakdown:** "We had 5 separate WhatsApp groups with 200+ members each. People forwarded the same rescue request 15 times with no timestamp. Volunteers sent food boats to the same temple 4 times while an interior pocket in Biyagama was cut off for two days without a single sip of clean water."

Conversation 5 -- Sri Lanka Red Cross Society (SLRCS) Branch Coordinator, Kalutara Branch

32 min · Phone interview

- **Context:** Coordinates volunteer disaster response teams, water bowsters, and mobile medical camps.

- **Audit and accountability gap:** *"After every flood, the government and donors ask: Why did you send this boat there? Who approved it? Without an immutable audit log, no one can prove that decisions were made fairly and based on objective priority rather than personal influence."*

Part 2: Quantitative User Validation Survey

32 respondents · Administered 22-25 September 2026 · Disaster volunteers, EMTs, local government staff, community leaders, flood survivors (Colombo & Gampaha districts)

Survey Question	Result	Operational Implication
What is your biggest obstacle coordinating disaster relief during peak floods?	93.8% (30/32) selected information chaos, duplicate requests & unverified rumours	Confirms need for automated schema extraction and spatial deduplication
Do you have real-time visibility into which rescue vehicles/boats are available?	87.5% (28/32): No -- rely on voice phone calls and individual WhatsApp chats	Proves acute market need for a unified live resource registry
How long does manual call triage and note-taking take per incoming distress call?	Average: 4.8 minutes per call (range: 3-10 min)	ResQGrid AI's ~2-second triage represents a >10x speedup
Have response teams ever been dispatched onto flooded or impassable routes?	81.3% (26/32): Yes, frequently	Validates core value of hazard route penalties and automatic re-planning
Would dispatchers trust an autonomous AI making life-or-death dispatch decisions?	96.9% (31/32): No -- a human commander must approve every dispatch	Directly justifies ResQGrid AI's motto: AI recommends; humans approve

Part 3: Verifiable Empirical Citations

- **World Bank Disaster Risk Profile -- Sri Lanka (2023/2024):** Confirms floods are Sri Lanka's most frequent natural hazard. Annualised damage exceeds \$313 million. Flood events documented annually for over 40 consecutive years.
- **BBC News & Sri Lanka DMC Situation Reports (November 2025):** 330+ fatalities, dozens missing in hill-country mudslides, widespread inundation across Western, Central, and Sabaragamuwa provinces.
- **UN OCHA / IOM Sri Lanka Situation Report (May 2016):** 301,000+ individuals displaced across 22 districts; Kelani River flooding severely paralysed the capital city's western corridor.
- **Sri Lanka Parliamentary Debrief on National Emergency Telephony (2024):** 117 Emergency Hotline documented call-drop rates exceeding 70% during peak monsoon events due to lack of concurrent digital intake lines.
- **Sri Lanka Red Cross Society Field Report (December 2024):** Documented duplicate food drops in accessible areas while interior Biyagama pockets remained unserved for 48+ hours -- attributed to absence of centralised real-time coordination data.
- **MarketsandMarkets -- Incident & Emergency Management Market Report (2024):** Global incident and disaster management systems market projected to reach \$135.8B by 2026, growing at 6.8% CAGR from \$107B in 2023.